



United International University
School of Business and Economics
BMT 2113: Business Mathematics II
QNT 2105: Mathematics for Business Decision
Summer 2026, Section: (A, B, D)

Instructor	: Mosabbir Ahmad
Class Hours	: Section A: Saturdays and Tuesdays, 09:51 am - 11:10 am (Room # 207) Section B: Sundays and Wednesdays, 08:30 am - 09:50 am (Room # 206) Section D: Sundays and Wednesdays, 12:31 pm - 01:50 pm (Room # 206)
Counseling Hours	: Saturdays and Tuesdays: 11:11 am - 01:50 pm Sundays and Wednesdays: 11:11 am - 12:30 pm
Email	: mosabbir@bus.uui.ac.bd (can't be used for seeking makeup exam permission)
Phone	: 01747954128
Office	: Room # 335 (1)

Course Objective:

- To guide you and enrich your acquaintance with the theories and practices of Business Mathematics.
- To help you improve analytical and computational skills necessary to face the challenges in the world of business.

Prerequisite: BMT 1103: Business Mathematics I

Required Materials:

Textbook: Mathematics with Applications in Management and Economics
- Gordon D. Prichett, John C. Saber, 7e, ISBN # 0-256-09237-0

Reference: Mathematical Applications for the Management, Life, and Social Sciences
- Tenth Edition, Ronald J. Harshbarger, James J. Reynolds

Performance Evaluation and Grading:

Your performance will be evaluated based upon the following criterion and your final (overall) grade will be determined on the following scale:

Assessment Plan:

Mid-Term Exam:	25%
Final	40%
Quizzes	15%
Home Assignment	15%
Attendance:	5%

Grading System:

A	≥90%
A-	86% - <90%
B+	82% - <86%
B	78% - <82%
B-	74% - <78%
C+	70% - <74%
C	66% - <70%
C-	62% - <66%
D+	58% - <62%
D	55% - <58%
F	<55%

Quizzes and Home Assignment:

The quizzes and home assignments are intended to keep you up-to-date with the materials already presented in the class. There will be three to four *unannounced and announced* quizzes, and several sets of home assignment problems. Quizzes will be based on the content from the lecture. Each quiz will comprise of numerical problems. The home assignments will primarily consist of end-of-chapter problems. You will surely observe the benefit of self-working on the home assignment problems on the quiz or term exam. Since your worst performed quiz will be dropped, **no make-up quiz will be taken.**

Also, no late submission of your assignments will be accepted. Please note that your scores on quizzes and homework will play a vital role on your final grade. So take it seriously.

Mid-Term and Final Exams:

The Mid-Term and the final exam will be designed to test your knowledge and understanding of the issues discussed in the classroom. Exams will be based on the content from the lecture. Each exam will consist of short questions, and numerical problems. No make-up exam will be administered except for extreme personal emergency. Under this situation, you have to seek prior permission from the course instructor over phone, inform the department, and are expected to provide appropriate written documentation. No email is accepted for seeking any permission for any makeup exam. However, 10% penalty will be imposed for any unauthorized absence.

Miscellaneous:

Schedule:	The schedule is subject to change at the discretion of the instructor.
Classroom Civility:	It is particularly important that a good climate for learning be provided by the instructor and students. Please join with the instructor to help avoid any form of harassment in this class. Students are expected to assist in maintaining a classroom environment conducive to learning.
Academic honesty:	Copying work done by others, either in-class or out of class, is an act of academic dishonesty and will be prosecuted to the full extent allowed by University policy. Collaboration during quizzes and exams is forbidden. For more information on university policies regarding scholastic dishonesty, see University Student Rules.
Attendance:	Students attending all classes will be awarded with the allocated 5 points. Your total attendance score will be automatically calculated by the UCAM system. Unauthorized absences for 3 consecutive classes will however expel the student from the course.

Tentative Schedule

	*Introduction, reviews of the syllabus and class policies
	* Introduction to Linear Programming
	* Graphical Method of solving Linear Programming problems
Week 01-06	* Software Applications in solving Linear Programming problems
	* Introduction to Differential Calculus, * Definition of the Derivative
	* d/dx Notation, Delta method, and Rules of Operation, * Review Problems
	* Applications of Differential Calculus and Marginal Functions

Mid-Term Exam

	* Maxima and Minima of Functions
	* The First Derivative Test * The Second Derivative Test
	* Additional Topics and more applications of Differential Calculus
	* Partial Derivatives and Calculus of Functions of Two or More Variables
Week 07 - 12	* Maxima and Minima of Functions of Two or More Variables,
	* MPS and the Multiplier.
	* Introduction to Integral Calculus, * Antiderivatives: The Indefinite Integral
	* Area and Definite Integral, * The Area between Two Curves,
	* Interpretive Applications of Area,
	* Consumers' and Producers' Surplus
	* Review Problems

Final Exam

* The schedule is tentative and subject to change according to the class demand and out-side circumstances.

* Date of Final Exam will be announced/assigned as per the university calendar.